Software Requirements Specification (SRS)

IEEE-style requirements specification

Project: Pharmacy Expiry & Re-order Dispatch Engine

Version: 1.0 | Status: Student Submission Draft | Date: 12 October 2026

Prepared for: Software Engineering Mini-Project Deliverables – Part 1

Revision History

Version	Date	Author	Change
1.0	12-Oct-2026	Student	Initial submission draft

1. Introduction

1.1 Purpose

This SRS defines functional and non-functional requirements for the Pharmacy Expiry & Re-order Dispatch Engine. It describes how pharmacy staff record medicine batches, select stock using First-Expiry-First-Out (FEFO), monitor stock levels, and generate purchase orders.

1.2 Scope

The system supports batch-level inventory tracking, expiry-aware dispensing lists, stock threshold monitoring, and purchase-order preparation/dispatch. It is intended to reduce expired-stock dispensing and help staff replenish medicines before stockouts. It does not replace a pharmacy's clinical decision-making or an external supplier's ordering platform.

1.3 Intended Audience

Pharmacy clerks, inventory managers, developers, testers, project evaluators, and maintenance personnel.

1.4 Definitions and Acronyms

Term	Definition
FEFO	First-Expiry-First-Out; choose the valid batch with the nearest expiry date first.
Batch	A quantity of one medicine associated with a batch identifier and expiry date.
Reorder threshold	Configured minimum stock level that triggers replenishment review.
FR / NFR	Functional requirement / Non-functional requirement.

1.5 References

- IEEE-style SRS structure as required by the course deliverables.
- Project decisions recorded in the Pharmacy Expiry and Reorder System planning work.

2. Overall Description

2.1 Product Perspective

The system is a modular application with user interface, inventory/batch management, FEFO selection, reorder monitoring, purchase-order handling, and persistent storage. Supplier dispatch may be represented as a recorded workflow unless a real supplier integration is configured.

2.2 Product Functions

- Record medicine batch details and stock quantity.
- Display medicine inventory and batch status.
- Generate FEFO dispensing lists while excluding expired or zero-stock batches.
- Monitor stock against configured reorder thresholds.
- Create and record purchase orders for replenishment.

2.3 User Classes

Actor	Responsibilities
Pharmacy Clerk	Record batches, view inventory, request FEFO dispensing list.
Inventory Manager	Configure reorder thresholds, review low stock, create/dispatch purchase orders.

Inventory Supplier	Receive purchase-order information and fulfil replenishment orders.

2.4 Assumptions and Dependencies

- Users are authenticated and assigned appropriate roles.
- Medicine identifiers, expiry dates, quantities, and reorder thresholds are entered accurately.
- Supplier contact details are maintained by the pharmacy.
- The system clock and date format are configured correctly.

3. Specific Requirements

3.1 Functional Requirements

ID	Name	Requirement (shall statement)	Priority	Measurable acceptance criteria
FR-01	Batch Management	The system shall allow an authorized Pharmacy Clerk to record a medicine batch with medicine ID, batch ID, expiry date, and non-negative quantity.	High	Given valid required fields, save the batch and show its identifier; reject missing IDs, invalid dates, and negative quantity.
FR-02	Inventory View	The system shall display current stock by medicine and batch, including quantity and expiry status.	High	For a stored batch, displayed medicine/batch IDs and quantity match persisted values; expired batches are labelled expired.
FR-03	FEFO Selection	The system shall order eligible batches by earliest expiry date and exclude expired or zero-stock batches from the dispensing list.	High	Given multiple batches, eligible batches appear in ascending expiry order; expired and zero-quantity batches are absent.
FR-04	Reorder Monitoring	The system shall compare available stock with each medicine's configured reorder threshold and flag stock below the threshold.	High	When available stock is below threshold, the medicine appears in the low-stock list; at or above threshold it is not flagged.
FR-05	Purchase Order	The system shall allow an Inventory Manager to generate and record a purchase order for a flagged medicine, including medicine, quantity, supplier, and status.	Medium	A valid order receives a unique order ID and initial status; missing supplier or non-positive quantity is rejected.

3.2 Non-Functional Requirements

ID	Quality attribute	Requirement	Priority	Measurable acceptance criteria
NFR-01	Performance	For up to 5,000 stored batch records, inventory search and FEFO list generation shall complete within 2 seconds for 95% of	Medium	Run 100 representative requests; at least 95 complete within 2 seconds.

		requests on the agreed test machine.		
NFR-02	Security	The system shall restrict batch changes, threshold configuration, and purchase-order creation to authenticated users with the required role.	High	Unauthenticated requests are denied; a clerk cannot access manager-only actions; no password or secret is written to logs.

3.3 Security Objectives and Requirements

Objective	Security requirement
Protect inventory integrity	SEC-01: Validate all user-supplied IDs, dates, quantities, and thresholds; reject invalid values and log a non-sensitive validation event.
Restrict privileged actions	SEC-02: Require authentication and role-based authorization for manager-only reorder and purchase-order operations.
Protect sensitive information	SEC-03: Do not log credentials, access tokens, or unnecessary personal data; restrict access to application logs.
Maintain auditability	SEC-04: Record actor, timestamp, action, and affected record ID for batch and purchase-order changes.

3.4 Use Case Model

Actors: Pharmacy Clerk, Inventory Manager, Inventory Supplier. Core use cases: Record Medicine Batch; View Medicine Inventory; Generate FEFO Dispensing List; Monitor Reorder Threshold; Generate Purchase Order; Dispatch Purchase Order; Configure Reorder Threshold.

Relationships: Generate FEFO Dispensing List «include» Validate Batch Availability. Generate Purchase Order «extend» Monitor Reorder Threshold when stock is below the configured threshold. Inventory Manager performs threshold configuration and order creation; Supplier participates in order receipt/fulfilment.

Use-case diagram (PlantUML source; render using PlantUML or recreate in draw.io):

```
@startuml
left to right direction
actor "Pharmacy Clerk" as Clerk
actor "Inventory Manager" as Manager
actor "Inventory Supplier" as Supplier
rectangle "Pharmacy Expiry & Re-order System" {
  usecase "Record Medicine Batch" as UC1
  usecase "View Medicine Inventory" as UC2
  usecase "Generate FEFO Dispensing List" as UC3
  usecase "Validate Batch Availability" as UC4
  usecase "Monitor Reorder Threshold" as UC5
  usecase "Generate Purchase Order" as UC6
  usecase "Dispatch Purchase Order" as UC7
  usecase "Configure Reorder Threshold" as UC8
}
Clerk --> UC1
Clerk --> UC2
Clerk --> UC3
UC3 .> UC4 : <<include>>
Manager --> UC2
Manager --> UC5
Manager --> UC6
Manager --> UC7
Manager --> UC8
Supplier --> UC7
```

UC6 .> UC5 : <<extend>>
@enduml

4. External Interface Requirements

4.1 User Interface

The interface shall provide forms for batch entry, inventory and expiry views, FEFO lists, low-stock alerts, and purchase-order creation. Validation errors shall identify the field requiring correction without exposing internal exceptions.

4.2 Software Interfaces

The application shall persist medicine, batch, threshold, supplier, and purchase-order records in a database. Any supplier integration shall use an authenticated interface; if unavailable, dispatch status may be recorded manually.

5. Constraints and Traceability

The FEFO rule must never select an expired or zero-stock batch. Thresholds are configurable by an authorized Inventory Manager. The following mapping links requirements to design components and tests.

Requirement	Primary component	Verification
FR-01	Batch Management	TC-01, TC-02
FR-02	Inventory Service	TC-03
FR-03	FEFO Selector	TC-04, TC-05
FR-04	Reorder Monitor	TC-06
FR-05	Purchase Order Service	TC-07, TC-08
NFR-01	Inventory/FEFO services	TC-09
NFR-02 / SEC-02	Authentication and Authorization	TC-10

Appendix A: Use-Case Specification – Generate FEFO Dispensing List

Primary actor: Pharmacy Clerk. Preconditions: clerk is authenticated; medicine has recorded batches. Main flow: (1) Clerk selects a medicine. (2) System retrieves its batches. (3) System removes expired and zero-stock batches. (4) System sorts remaining batches by earliest expiry date. (5) System displays the eligible list and quantities. Alternate flow: if no eligible batch remains, the system displays “No eligible stock available” and does not recommend a batch. Postcondition: no inventory quantity is changed merely by viewing the list.